

Reconstruction from Observation: The Density Bridge and Measure-Theoretic Takens

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Abstract

Given a measure-preserving system $(X, \mathcal{B}, \mu, T)$ and a bounded observation function $h \in L^\infty(X, \mu)$, the time series $h(x), h(Tx), h(T^2x), \dots$ is all an observer sees. The question is when this is enough: when can the observer recover the full state space X from the time-delayed record?

The answer is a three-way equivalence. State-space recovery — $\mathcal{O}_h := \sigma(\{h \circ T^n : n \in \mathbb{Z}\}) = \mathcal{B}$ modulo μ — holds if and only if the algebra generated by the delay orbit $\{h \circ T^n\}$ is dense in $L^2(X, \mu)$, if and only if the delay map $\Phi_h : x \mapsto (h(T^n x))_{n \in \mathbb{Z}}$ is a measure-theoretic embedding $X \hookrightarrow \mathbb{R}^{\mathbb{Z}}$.

The equivalence of the first two conditions is the *density bridge*: for any subalgebra $\mathcal{A} \subseteq L^\infty$ containing the constant 1, $\overline{\mathcal{A}}^{L^2} = L^2(X, \sigma(\mathcal{A}), \mu)$. The proof is an instance of the functional monotone class theorem.

The cyclic vector condition $\overline{\text{span}}\{U_T^n h\} = L^2$ implies reconstruction but is strictly stronger: it is a spectral condition on the Koopman operator U_T , while reconstruction is a condition on the Boolean algebra of observable events. They coincide precisely when U_T has simple L^2 -spectrum.

When reconstruction holds, the compact Stone space of the observable algebra — introduced in [3] as a technical device for deriving σ -additivity from compactness — is measure-theoretically isomorphic to X itself. Compared to the classical Takens embedding theorem [4], the reconstruction theorem requires only measurability and uses all delays simultaneously; the algebraic density condition replaces the dimension count.

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1 Introduction

An observer of a measure-preserving system $(X, \mathcal{B}, \mu, T)$ has access to a bounded observation function $h : X \rightarrow \mathbb{R}$, and from a single initial point x sees the time series $h(x), h(Tx), h(T^2x), \dots$. The prior papers established that coherent structured observations determine a probability measure and that measure determines the dynamics. The question remaining is whether the observer can recover the state space itself: whether the orbit of measurements uniquely identifies where in X they are.

The answer depends on how much the delay orbit $\{h \circ T^n : n \in \mathbb{Z}\}$ collectively distinguishes points of X . If h is constant, no reconstruction is possible. At the other extreme, if the observable σ -algebra $\mathcal{O}_h = \sigma(\{h \circ T^n\})$ equals $\mathcal{B}(X)$ modulo μ , then almost every pair of distinct points is separated by some delayed measurement. The reconstruction theorem shows this σ -algebraic condition is equivalent to two others: an analytic condition on L^2 -approximation by the delay algebra, and a geometric condition on the delay map $\Phi_h : x \mapsto (h(T^n x))_{n \in \mathbb{Z}}$ being a measure-theoretic embedding.

The key to the equivalence is a general lemma on function algebras — the density bridge — which says that for any subalgebra $\mathcal{A} \subseteq L^\infty(X, \mu)$ containing 1, the L^2 -closure of $\mathcal{A}$ equals

$L^2(X, \sigma(\mathcal{A}), \mu)$. The proof goes via the functional monotone class theorem: the class of bounded measurable functions approximable in L^2 by elements of $\mathcal{A}$ is a vector space closed under bounded monotone limits, and since $\mathcal{A}$ is an algebra it already separates points of $\sigma(\mathcal{A})$, so the class is all of $L^\infty(\sigma(\mathcal{A}))$. The σ -algebra and the L^2 space it generates are inextricably linked.

The Koopman operator $U_T f = f \circ T$ on $L^2(X, \mu)$ provides a spectral perspective. The cyclic vector condition — $\overline{\text{span}}\{U_T^n h : n \in \mathbb{Z}\} = L^2$ — implies reconstruction but is strictly stronger: it asks for density of the *linear span* of the orbit, while reconstruction asks only for density of the *algebra* the orbit generates. For bounded h , the algebra contains the linear span, so cyclicity is the stronger condition. The two coincide precisely when U_T has simple L^2 -spectrum.

When reconstruction holds, a compact space introduced earlier for an entirely different purpose — the Stone space $\text{St}(\mathcal{O}_h)$ of the observable algebra, built in [3] as a technical completion of the sample space for deriving σ -additivity — turns out to be measure-theoretically isomorphic to X itself. The construction that began by seeking a compact ambient space for the measure ends by identifying it with the state space being observed.

Compared to the classical Takens embedding theorem [4] — which requires a C^2 diffeomorphism on a compact d -manifold and a generic smooth h , and produces a diffeomorphic embedding into $\mathbb{R}^{2d+1}$ — the reconstruction theorem here requires only measurability, uses all finite delays simultaneously, and replaces the dimension count with the algebraic density condition. On a compact manifold, Stone–Weierstrass gives density for any point-separating h , recovering the measure-theoretic content of Takens without differentiability hypotheses.

Throughout, $(X, \mathcal{B}, \mu)$ is a standard probability space and $T : X \rightarrow X$ is a measurable, μ -preserving invertible bimeasurable bijection with $T_*\mu = \mu$. We fix $h \in L^\infty(X, \mu)$.

Definition 1 (Observable algebra). The *observable algebra* generated by h is

$$\mathcal{O}_h := \sigma(\{h \circ T^n : n \in \mathbb{Z}\}) \subseteq \mathcal{B}(X).$$

The *delay algebra* $\mathcal{A}_h$ is the smallest subalgebra of $L^\infty(X, \mu)$ containing 1 and every $h \circ T^n$; concretely, finite sums $\sum_\alpha c_\alpha \prod_k (h \circ T^{n_k})^{e_k}$.

Definition 2 (Delay map). The *delay map* is

$$\Phi_h : X \rightarrow \mathbb{R}^{\mathbb{Z}}, \quad \Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}.$$

Since $\pi_n \circ \Phi_h = h \circ T^n$ for each coordinate projection π_n , the map Φ_h is measurable and $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}}))$. Thus $\mathcal{O}_h = \mathcal{B} \bmod \mu$ is precisely the condition that Φ_h is a measure-theoretic embedding — the geometric face of condition (iii) in the reconstruction theorem below.

2 Density bridge and reconstruction

The central lemma connects the Boolean algebra level to the L^2 level. It holds for any subalgebra of L^∞ .

Lemma 3 (Density bridge). *Let $(X, \mathcal{B}, \mu)$ be a probability space and let $\mathcal{A} \subseteq L^\infty(X, \mu)$ be a subalgebra containing the constant function 1. Then:*

$$\overline{\mathcal{A}}^{L^2} = L^2(X, \sigma(\mathcal{A}), \mu).$$

Consequently, $\overline{\mathcal{A}}^{L^2} = L^2(X, \mu)$ if and only if $\sigma(\mathcal{A}) = \mathcal{B}$ modulo μ .

Proof. Write $\mathcal{M} := \sigma(\mathcal{A})$ and $V := \overline{\mathcal{A}}^{L^2}$.

Step 1: $V \subseteq L^2(X, \mathcal{M}, \mu)$. Every $f \in \mathcal{A}$ is $\mathcal{A}$ -measurable, hence $\mathcal{M}$ -measurable. So $\mathcal{A} \subseteq L^2(X, \mathcal{M}, \mu)$. Since $L^2(X, \mathcal{M}, \mu)$ is a closed subspace of $L^2(X, \mathcal{B}, \mu)$, taking the closure gives $V \subseteq L^2(X, \mathcal{M}, \mu)$.

Step 2: $L^2(X, \mathcal{M}, \mu) \subseteq V$. It suffices to show that every indicator $\mathbf{1}_E$ with $E \in \mathcal{M}$ lies in V . By the functional monotone class theorem [1], the class

$$\mathcal{H} := \{f \in L^\infty(X, \mathcal{M}, \mu) : f \in V\}$$

is a vector space containing $\mathcal{A}$ and closed under bounded monotone pointwise limits (since if $f_n \in V$ are uniformly bounded and $f_n \rightarrow f$ pointwise a.e. with $f \in L^\infty$, then $f_n \rightarrow f$ in L^2 by dominated convergence, so $f \in V$). Since $\mathcal{A}$ is an algebra containing 1 and $\sigma(\mathcal{A}) = \mathcal{M}$, the monotone class theorem gives $\mathcal{H} = L^\infty(X, \mathcal{M}, \mu)$. In particular every $\mathcal{M}$ -indicator $\mathbf{1}_E \in V$.

Since the $\mathcal{M}$ -simple functions are dense in $L^2(X, \mathcal{M}, \mu)$ and V is closed, $L^2(X, \mathcal{M}, \mu) \subseteq V$.

Conclusion. Steps 1 and 2 give $V = L^2(X, \mathcal{M}, \mu)$. The final equivalence is immediate: $V = L^2(X, \mathcal{B}, \mu)$ iff $L^2(X, \mathcal{M}, \mu) = L^2(X, \mathcal{B}, \mu)$ iff $\mathcal{M} = \mathcal{B} \bmod \mu$. $\square$

Remark 4 (Role of the algebra structure). The algebra structure of $\mathcal{A}$ is essential. The corresponding statement for the closed linear span $\mathcal{H}_{\mathcal{A}} = \overline{\text{span}}(\mathcal{A})$ is false in general: $\mathcal{H}_{\mathcal{A}} = L^2$ does not imply $\sigma(\mathcal{A}) = \mathcal{B}$.

For example, take $X = [0, 1]$, $\mu = \text{Lebesgue}$, $T(x) = 2x \bmod 1$ (the doubling map), and $h(x) = x$. The orbit $\{h \circ T^n\}(x) = \{2^n x \bmod 1\}$ generates $\mathcal{B}([0, 1])$ as a σ -algebra (the dyadic structure separates points). But $\text{span}\{2^n x \bmod 1 : n \geq 0\}$ is a proper subspace of $L^2[0, 1]$: it does not contain x^2 . So the linear span can be a proper dense subspace even when $\sigma(\mathcal{A}) = \mathcal{B}$.

In the other direction, h cyclic for U_T (linear span dense) always implies $\sigma(\mathcal{A}_h) = \mathcal{B} \bmod \mu$ — see Corollary 8.

Theorem 5 (Reconstruction theorem). *Let $(X, \mathcal{B}, \mu, T)$ be an invertible measure-preserving system and $h \in L^\infty(X, \mu)$. The following are equivalent:*

- (i) $\mathcal{O}_h = \mathcal{B} \bmod \mu$;
- (ii) $\mathcal{A}_h$ is dense in $L^2(X, \mu)$;
- (iii) $\Phi_h : X \rightarrow \mathbb{R}^{\mathbb{Z}}$ is a measure-theoretic embedding: $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}})) = \mathcal{B} \bmod \mu$.

Proof. (i) $\Leftrightarrow$ (ii). Apply the density bridge (Lemma 3) with $\mathcal{A} = \mathcal{A}_h$ and $\sigma(\mathcal{A}_h) = \mathcal{O}_h$: $\overline{\mathcal{A}_h}^{L^2} = L^2(X, \mu)$ iff $\mathcal{O}_h = \mathcal{B} \bmod \mu$.

(i) $\Leftrightarrow$ (iii). By Definition 2, $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}}))$. So $\mathcal{O}_h = \mathcal{B} \bmod \mu$ iff $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}})) = \mathcal{B} \bmod \mu$, which is exactly condition (iii). $\square$

Remark 6 (What “measure-theoretic embedding” means). Condition (iii) says that Φ_h is *almost everywhere injective* and the pullback σ -algebra is all of $\mathcal{B}$. More precisely: since $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}})) = \mathcal{B} \bmod \mu$, for any two sets $A, A' \in \mathcal{B}$ with $\mu(A \triangle A') > 0$ there exists a Borel set $E \subseteq \mathbb{R}^{\mathbb{Z}}$ with $\mu(\Phi_h^{-1}(E) \triangle A) = 0$ and $\mu(\Phi_h^{-1}(E) \triangle A') > 0$. In measure-theoretic language, Φ_h induces an isomorphism of measure algebras $(X/\mu, \mathcal{B}/\mu) \cong (\mathbb{R}^{\mathbb{Z}}/(\Phi_h)_*\mu, \mathcal{B}(\mathbb{R}^{\mathbb{Z}})/(\Phi_h)_*\mu)$.

The T -invariance of μ implies that $(\Phi_h)_*\mu$ is invariant under the bilateral shift σ on $\mathbb{R}^{\mathbb{Z}}$, since $(\Phi_h(Tx))_n = h(T^{n+1}x) = (\sigma(\Phi_h(x)))_n$ for each n . The dynamical system (X, T, μ) is therefore isomorphic to $(\mathbb{R}^{\mathbb{Z}}, \sigma, (\Phi_h)_*\mu)$ restricted to the image of Φ_h .

Definition 7 (Cyclic vector). The function $h \in L^2(X, \mu)$ is a *cyclic vector* for U_T if

$$\mathcal{H}_h := \overline{\text{span}}\{U_T^n h : n \in \mathbb{Z}\} = L^2(X, \mu).$$

Corollary 8 (Cyclic implies reconstruction). *If $h \in L^\infty(X, \mu)$ is a cyclic vector for U_T , then $\mathcal{O}_h = \mathcal{B} \bmod \mu$.*

Proof. Since $\mathcal{A}_h$ contains $\text{span}\{h \circ T^n\} = \text{span}\{U_T^n h\}$ as a subset, $\overline{\mathcal{A}_h}^{L^2} \supseteq \mathcal{H}_h = L^2$. By the density bridge, $\mathcal{O}_h = \mathcal{B} \bmod \mu$. $\square$

Remark 9 (The implication is strict). The converse of Corollary 8 fails in general: $\mathcal{O}_h = \mathcal{B}$ does not imply h is cyclic. The doubling map example of Remark 4 witnesses this.

The cyclic vector condition asks for density of the *linear span* of the orbit; reconstruction asks only for density of the *algebra*. For a bounded h , the linear span is contained in the algebra, so cyclicity is the stronger condition.

The two conditions coincide precisely when U_T has *simple L^2 -spectrum*: $L^2(X, \mu)$ is a cyclic U_T -module, meaning there exists some cyclic vector. In this case, every generating observation h (satisfying $\mathcal{O}_h = \mathcal{B}$) is automatically cyclic. Simple spectrum is a generic property of ergodic transformations in the Baire category sense [2] but is not universal.

The Stone space $\text{St}(\mathcal{O}_h)$ of the observable algebra is the compact Hausdorff space whose points are ultrafilters on $\mathcal{O}_h$, with the natural Stone embedding $\iota : X \rightarrow \text{St}(\mathcal{O}_h)$ sending x to the principal ultrafilter $\{E \in \mathcal{O}_h : x \in E\}$.

Proposition 10 (Stone space identification). *Suppose $\mathcal{O}_h = \mathcal{B}$ modulo μ . Then ι is a measure-theoretic isomorphism: ι identifies the measure algebra $(X/\mu, \mathcal{B}/\mu)$ with $(\text{St}(\mathcal{O}_h)/\iota_*\mu, \mathcal{B}(\text{St}(\mathcal{O}_h))/\iota_*\mu)$.*

Proof. Injectivity mod μ . Suppose $\iota(x) = \iota(x')$. For any $E \in \mathcal{B}$ with $x \in E \not\equiv x' \pmod{\mu}$, find $F \in \mathcal{O}_h$ with $\mu(E \triangle F) = 0$. Since $x \in F$ and $\iota(x) = \iota(x')$, also $x' \in F$, so $x' \in (E \triangle F) \cup F^c$, placing x' in a null set. Hence ι is injective μ -a.e.

σ -algebra identification. The pullback ι^{-1} maps the Baire σ -algebra of $\text{St}(\mathcal{O}_h)$ (generated by the clopen sets $[E] = \{u : E \in u\}$) to $\mathcal{O}_h = \mathcal{B} \pmod{\mu}$.

Measure recovery. Since μ is σ -additive on $\mathcal{O}_h = \mathcal{B}$, the collective exhaustion condition of [3] forces $\iota_*\mu$ to concentrate on the principal ultrafilters $\iota(X)$, so $\iota_*\mu$ is supported on the image and the measure algebras are identified. $\square$

The Stone space introduced in [3] as a compact ambient space for deriving σ -additivity is, under the reconstruction condition, the state space X itself. The construction that began by seeking a compact completion of the sample space ends by recognising it as the object the observer was trying to recover.

The classical Takens embedding theorem [4] reaches a related conclusion by a different route: for a C^2 diffeomorphism on a compact d -manifold and a generic smooth h , the finite-delay map $\Phi_h^{(k)}(x) = (h(x), h(Tx), \dots, h(T^{k-1}x))$ is a diffeomorphic embedding into $\mathbb{R}^{2d+1}$ for $k \geq 2d + 1$. The reconstruction theorem here requires only measurability, uses all delays simultaneously, and replaces the dimension count with the algebraic density condition; in exchange, the embedding is measure-theoretic rather than topological. On a compact manifold with generic smooth h , the orbit $\{h \circ T^n\}$ separates points, so Stone–Weierstrass gives $\mathcal{A}_h$ uniformly dense in $C(M)$ and hence L^2 -dense: the density condition holds and the two results are compatible.

The question the theorem leaves open is the genericity of the reconstruction condition in the purely measure-theoretic setting. On compact manifolds the analysis is classical; on abstract standard probability spaces, characterising the pairs (T, h) for which $\mathcal{O}_h = \mathcal{B}$ modulo μ — and understanding how the answer varies with the spectral properties of T — is the natural next problem.

References

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